ORBITS OF $\mathrm{Aut}_{\mathbb{F}_5}(X_1(42))$ ON PLACES OF SMALL DEGREE: MATHEMATICS AND COMPUTATIONAL DETAILS

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Abstract. This note documents the computation of the orbits of the automorphism group $\mathrm{Aut}_{\mathbb{F}_5}(X) = \langle \sigma_4 \rangle$ of $X = X_1(42)/\mathbb{F}_5$ on the places of degree $1, \ldots, 5$, and explains how the result is stored and reused. Every step is exact. The image of a place is found from the pull-back σ^* alone, with no need to compute the inverse automorphism or any point map. The computation is split into independent parallel jobs, whose results are merged and checked. The note is a companion to the scripts `X1_42_orbits.m`, `X1_42_orbits_merge.py`, `run_orbits.sh`, `X1_42_orbits_load.m` and `X1_42_orbits_check.m`; it is not part of the manuscript.

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1. Setup

1.1. **The curve.** Let $X = X_1(42)$ over $\mathbb{F}_5$; it has genus 25. We use the plane model from Sutherland's tables (file `FFFc42.txt`). Its function field is

$$K = \mathbb{F}_5(X) = \mathbb{F}_5(t)[y]/(F(t,y)), \qquad [K : \mathbb{F}_5(t)] = 16,$$

realised in Magma as `AFF := AlgorithmicFunctionField(FunctionField(XF2))`, with $t =$ `BaseField(AFF).1` and $y =$ `AFF.1`. All places are places of $K/\mathbb{F}_5$. For $v \geq 1$ let $\mathcal{P}_v$ be the list

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`Places(AFF,v)` of places of degree v, in Magma's order, and write $P_i^{(v)}$ for its i-th entry. The sizes are

v	1	2	3	4	5	6	7	8	9
$\#\mathcal{P}_v$	12	6	8	183	648	2288	11160	48708	214384

The present computation covers $v \le 5$.

1.2. **The group.** By the computation recorded in `X1_42_automorphisms.tex`, $G = \mathrm{Aut}_{\mathbb{F}_5}(X)$ is cyclic of order 12, generated by $\sigma = \sigma_4$. The group is stored in `X1_42_aut_F5.txt` as the pairs $(\sigma^* t, \sigma^* y) \in K^2$ of pull-backs. The scripts never recompute the group: they read the file with `LoadAutDataOpt(name,false)`, which takes about 2 s and skips the closure test. They then check that σ_4 has order exactly 12, by composing σ_4^* with itself until the identity (t, y) appears; this takes about 87 s. On the rational places σ_4 acts as the permutation

$$A_4 = [9, 5, 2, 1, 12, 8, 11, 7, 3, 4, 10, 6], \tag{1}$$

which is the 12-cycle $(1\,9\,3\,2\,5\,12\,6\,8\,7\,11\,10\,4)$. This was found independently by `X1_42_check_orbs.m`.

2. Automorphisms acting on places

2.1. **From the pull-back to places.** Let $\phi \in G$ and write $\phi^* \colon K \to K$ for the corresponding $\mathbb{F}_5$-automorphism of the function field. For a place P with valuation ring $\mathcal{O}_P$, put

$$\mathcal{O}_{\phi(P)} = \{f \in K : \ \phi^* f \in \mathcal{O}_P\} = (\phi^*)^{-1}(\mathcal{O}_P).$$

Since ϕ^* is a field automorphism fixing $\mathbb{F}_5$, this is again a valuation ring of $K/\mathbb{F}_5$. It defines a place $\phi(P)$.

Lemma 2.1. *For every place P and every $f \in K$:*

(a) $v_{\phi(P)}(f) = v_P(\phi^* f)$*;*
(b) ϕ^* *induces an $\mathbb{F}_5$-isomorphism $\bar{\phi} \colon \kappa(\phi(P)) \to \kappa(P)$ of residue fields, with $\bar{\phi}(f(\phi(P))) = (\phi^* f)(P)$ whenever $f \in \mathcal{O}_{\phi(P)}$. In particular* $\deg \phi(P) = \deg P$*;*
(c) $P \mapsto \phi(P)$ *is a bijection on the places of each degree v, and $\phi \mapsto (P \mapsto \phi(P))$ is an action of G, possibly through the opposite group, which does not change the orbits.*

Proof. (a) holds because $v_P \circ \phi^*$ is a normalised discrete valuation of K with valuation ring $\mathcal{O}_{\phi(P)}$. For (b), ϕ^* maps $\mathcal{O}_{\phi(P)}$ onto $\mathcal{O}_P$ and, by (a), maps the maximal ideal onto the maximal ideal. It fixes $\mathbb{F}_5$, so the induced map of residue fields is an $\mathbb{F}_5$-isomorphism, and the degree is preserved. For (c), $(\phi^{-1})^*$ provides the inverse map, and $(\phi\psi)^* = \psi^*\phi^*$ shows that composition is respected up to order. □

Because $G = \langle \sigma \rangle$ is cyclic, the G-orbit of a place P is $\{\sigma^k(P) : 0 \le k < 12\}$. Let π_v be the permutation of $\{1, \dots, \#\mathcal{P}_v\}$ with $\sigma(P_i^{(v)}) = P_{\pi_v(i)}^{(v)}$. Then:

Corollary 2.2. *The G-orbits on $\mathcal{P}_v$ are exactly the cycles of π_v. Every cycle length divides* 12*, and $\pi_v^{12} = \mathrm{id}$.*

So it suffices to compute the single permutation π_v for each degree v. The difficulty is that σ is known only through σ^*. Computing $\sigma(P)$ directly as a place would require the ideal $(\sigma^*)^{-1}(\mathcal{O}_P)$. Instead, we *identify* $\sigma(P)$ inside the already computed list $\mathcal{P}_v$, using only valuations and residues of explicit functions at P. By Lemma 2.1, these are values at $\sigma(P)$ in disguise.

3. Identifying the image of a place

Fix a degree v and a place $P \in \mathcal{P}_v$. The image $\sigma(P)$ is found in at most three exact steps. Each step shrinks a set of candidates that is *guaranteed* to contain $\sigma(P)$.

3.1. **Encoding polynomials.** For a polynomial $g = \sum_{i=0}^{n} c_i x^i \in \mathbb{F}_5[x]$ with $c_n \neq 0$, put

$$\mathrm{Enc}(g) = \sum_{i=0}^{n} c_i 5^i + 5^{n+1} \in \mathbb{Z}_{>0}, \qquad c_i \in \{0, \dots, 4\}.$$

Enc is injective: the leading power 5^{n+1} recovers n, and the base-5 digits below it recover the coefficients. For α in a finite extension of $\mathbb{F}_5$, let $\mathrm{mp}(\alpha) \in \mathbb{F}_5[x]$ denote its minimal polynomial over $\mathbb{F}_5$.

3.2. **Step 1: the bucket.** The place of $\mathbb{F}_5(t)$ under a place Q of K is determined by $\mathrm{mp}(t(Q))$ if t is regular at Q, and is the infinite place if t has a pole at Q. Define the *bucket* of Q as

$$b(Q) = \begin{cases} \mathrm{Enc}(\mathrm{mp}(t(Q))) & v_Q(t) \geq 0, \\ 0 & v_Q(t) < 0. \end{cases}$$

We compute $b(Q)$ for all $Q \in \mathcal{P}_v$ once, and group the indices by bucket.

Lemma 3.1. *Put $b^\sigma(P) = \mathrm{Enc}(\mathrm{mp}((\sigma^* t)(P)))$ if $v_P(\sigma^* t) \geq 0$, and $b^\sigma(P) = 0$ otherwise. Then $b(\sigma(P)) = b^\sigma(P)$.*

Proof. By Lemma 2.1(a), $v_{\sigma(P)}(t) = v_P(\sigma^* t)$, so t has a pole at $\sigma(P)$ iff $\sigma^* t$ has a pole at P. Otherwise $\bar{\sigma}(t(\sigma(P))) = (\sigma^* t)(P)$ by Lemma 2.1(b). An $\mathbb{F}_5$-isomorphism of fields preserves minimal polynomials over $\mathbb{F}_5$. □

So the first candidate set is $C_1 = \{j : \ b(P_j^{(v)}) = b^\sigma(P)\} \subseteq \mathcal{P}_v$. It contains $\pi_v(i)$ for $P = P_i^{(v)}$, and has at most 16 elements, since $\sum e_i f_i = [K : \mathbb{F}_5(t)] = 16$ over any place of $\mathbb{F}_5(t)$. If $\#C_1 = 1$, the image is found.

3.3. **Step 2: the key.** For a place Q at which t and y are both regular, put $a = t(Q)$ and $c = y(Q)$, and define its *key*

$$\kappa(Q) = \sum_{m=0}^{3} \mathrm{Enc}(g_m)\,(5^{12})^{3-m}, \qquad (g_0, g_1, g_2, g_3) = \big(\mathrm{mp}(a), \mathrm{mp}(c), \mathrm{mp}(a+c), \mathrm{mp}(a+2c)\big).$$

If t or y has a pole at Q, we set $\kappa(Q) = 0$, meaning "not available". All g_m have degree at most $v \leq 9$, so $\mathrm{Enc}(g_m) < 5^{11} < 5^{12}$, and κ is an injective function of the quadruple $(g_0, \dots, g_3)$.

Lemma 3.2. *If $\sigma^* t$ and $\sigma^* y$ are regular at P, then $\kappa(\sigma(P))$ equals the key computed from the pair $\big((\sigma^* t)(P), (\sigma^* y)(P)\big)$ in place of (a, c).*

Proof. Use the same isomorphism $\bar{\sigma}$ for both coordinates. It sends $(t(\sigma(P)), y(\sigma(P)))$ to $((\sigma^* t)(P), (\sigma^* y)(P))$, and hence sends $a + c$ and $a + 2c$ to the corresponding combinations. Minimal polynomials over $\mathbb{F}_5$ are preserved. □

The second candidate set is $C_2 = \{j \in C_1 : \ \kappa(P_j^{(v)}) = \kappa^\sigma(P)\}$. It still contains the true image. If $\#C_2 = 1$, the image is found. If C_2 is empty (which can only happen when a key is unavailable), or if $\kappa^\sigma(P)$ cannot be computed, we continue with $C = C_1$; otherwise we continue with $C = C_2$.

Remark 3.3. Two different places can have the same key: for example, two places over the same place of $\mathbb{F}_5(t)$ whose residues (a, c) are conjugate over $\mathbb{F}_5$ only coordinatewise, but not as a pair. The four minimal polynomials make such collisions rare, and the step is cheap. Step 3 handles whatever remains.

3.4. **Step 3: the uniformizer pattern.** Let $C = \{j_1, \dots, j_r\}$ with $r \geq 2$, and write $Q_s = P_{j_s}^{(v)}$. For every candidate Q_s, choose a function $u_s \in K$ with $v_{Q_s}(u_s) > 0$. In practice $u_s =$ `LocalUniformizer`(Q_s), and the script checks $v_{Q_s}(u_s) > 0$. Nothing is required of u_s at the other candidates. Record:

- the zero pattern $Z_s = \{q \in \{1, \dots, r\} : \ v_{Q_q}(u_s) > 0\}$, which always contains s;
- the test bit $\beta_s = \big[\, v_P(\sigma^* u_s) > 0 \,\big] \in \{0, 1\}$.

A candidate q *survives* if $q \in Z_s \iff \beta_s = 1$ for every $s = 1, \dots, r$.

Proposition 3.4. *Let q^* be the index with $Q_{q^*} = \sigma(P)$.*

(a) *q^* always survives.*
(b) *If exactly one candidate survives, it is q^*.*
(c) *A candidate $q \neq q^*$ survives only if u_q vanishes at $\sigma(P)$.*

Proof. By Lemma 2.1(a), $\beta_s = [v_{\sigma(P)}(u_s) > 0] = [q^* \in Z_s]$, which proves (a). (b) follows from (a). For (c): $q \in Z_q$, so q can survive only if $\beta_q = 1$, i.e. $v_{\sigma(P)}(u_q) > 0$. □

So Step 3 is never wrong; at worst it is inconclusive. In that case the image is recorded as -1 (*ambiguous*), and the whole degree is rejected at the merge stage (Section 5). Ambiguity is unlikely: by (c), it requires the chosen uniformizer of another candidate to vanish at the true image. Should it ever occur, it can be removed by replacing u_q by another function vanishing at Q_q but not at the competing candidates (e.g. $u_q + \lambda u_q^2$ or a weak-approximation element), or by the constraint that π_v is a bijection.

Example 3.5. Suppose $C = \{1, 2, 3\}$, with zero patterns $Z_1 = \{1\}$, $Z_2 = \{2, 3\}$ and $Z_3 = \{3\}$, and observed bits $(\beta_1, \beta_2, \beta_3) = (0, 1, 1)$. Candidate 1 fails at $s = 1$, since $1 \in Z_1$ but $\beta_1 = 0$. Candidate 2 fails at $s = 3$, since $2 \notin Z_3$ but $\beta_3 = 1$. Candidate 3 satisfies all three conditions. Hence $\sigma(P) = Q_3$. With bits $(0, 1, 0)$ instead, only candidate 2 survives.

3.5. **Cost.** The functions $\sigma^* t$, $\sigma^* y$ and $\sigma^* u_s = u_s(\sigma^* t, \sigma^* y)$ (`ApplyAut`) are exact elements of K. Steps 1 and 2 need two residue evaluations and four minimal polynomials per place. Step 3 needs at most 16 local uniformizers, 16^2 valuations at the candidates, and 16 valuations of pulled-back functions at P. The pulled-back functions are the expensive part, which is why Step 3 is used only when Steps 1–2 leave several candidates. The script reports, for each degree, how many images were decided by each step.

4. Checks

The following are verified before a degree is saved.

(C1) *Group.* The loaded σ_4 has order exactly 12.
(C2) *Consistent numbering.* All parallel jobs for a degree use the same list $\mathcal{P}_v$. Each job writes $\#\mathcal{P}_v$ and the checksum

$$\mathrm{CHK} = \Big(\cdots \big((0 \cdot B + h_1)\, B + h_2 \big) \cdots \Big) \bmod M, \qquad h_i = \big(b_i \bmod p \big) + 7 \big(\kappa_i \bmod p \big),$$

with $B = 1000003$, $p = 10^9 + 7$, $M = 2^{61} - 1$, and b_i, κ_i the bucket and key of $P_i^{(v)}$. The merge requires all parts to agree.
(C3) *Completeness.* Every index has exactly one image, computed by the job responsible for it, and no image is ambiguous.
(C4) *Permutation.* The images form a permutation of $\{1, \dots, \#\mathcal{P}_v\}$. By Lemma 2.1(c), this must hold, so it is a non-trivial check of the whole identification.
(C5) *Group law.* All cycle lengths divide 12 (Corollary 2.2).

(C6) *Known case.* For $v = 1$, the permutation equals A_4 in (1) or its inverse. The inverse is accepted because the convention $\phi \mapsto \phi(P)$ versus $\phi \mapsto \phi^{-1}(P)$ only replaces π_v by π_v^{-1}, with the same orbits.

If any check fails for a degree, no file is written for that degree, and an old file of that degree is deleted, so a stale result can never be reused.

5. Parallel computation

5.1. **Splitting.** The images of different places are independent, so the work splits trivially. With m parts for degree v, job $k \in \{0, \dots, m-1\}$ computes $\pi_v(i)$ exactly for

$$i \equiv k + 1 \pmod{m}.$$

This round-robin split balances the load better than contiguous blocks, because Magma's order of $\mathcal{P}_v$ groups places by the place of $\mathbb{F}_5(t)$ below them, and bucket sizes vary.

Each job still computes the full list $\mathcal{P}_v$ and the buckets and keys of *all* places, since any of them may be a candidate. It also loads the group and checks the order. These shared costs are small compared with the image computations for $v = 4, 5$.

5.2. **Launch and merge.** `run_orbits.sh P1 P2 P3 P4 P5` starts P_v jobs for degree v; the default is $1, 1, 1, 4, 12$, i.e. 19 jobs, one per core:

`magma -b degs:=`v `part:=`k `nparts:=`m `X1_42_orbits.m`

Job (v, k, m) writes `X1_42_orbits_d<v>_part<k>_of_<m>.txt`, which contains:

- a header with `DEG`, `GEN`, `PART`, `NPARTS`, `COUNT` and `CHECKSUM`;
- for part 0 only, the place data lines `P i | c0 c1 ... | key`;
- its lines `I i j`, meaning $\pi_v(i) = j$;
- a closing `END` line.

After all jobs finish, `X1_42_orbits_merge.py` performs checks (C2)–(C6), splits π_v into cycles, and writes the final files.

6. The saved data

For each degree v, the file `X1_42_orbits_d<v>.txt` contains:

line	meaning
`X1_42_ORBITS v1 DEG v GEN 4 COUNT n ORBITS k CHECKSUM c`	header, continued by
`P i | c0 c1 ... cn | key`	place i: coefficients of $\mathrm{mp}(t(P_i))$, constant term first (empty for $t = \infty$), and $\kappa(P_i)$
`I i j`	$\sigma_4(P_i) = P_j$
`O k s : i1 i2 ... is`	orbit k of size s, as the cycle $i_1 \mapsto i_2 \mapsto \cdots$
`R k i`	representative of orbit k (its smallest index)
`END n`	end marker

`X1_42_orbits_summary.txt` lists, for each degree, the number of orbits and the distribution of orbit sizes.

6.1. **Reuse.** `X1_42_orbits_load.m` provides three functions.

- `LoadOrbitFile(v)` reads the file into a record with fields `coeffs`, `keys`, `img`, `orbits`, `reps` and `orbitof`. It needs no curve computation.
- `PlacesFromOrbitFile(D)` recomputes $\mathcal{P}_v$ and verifies, place by place, the stored polynomial and key and the checksum. Afterwards, index i in the file is guaranteed to refer to `PL[i]` in the current session.

- `FindPlace(coeffs, key, v)` recovers a single place without computing $\mathcal{P}_v$. It takes the zeros of $p(t)$, where p is the stored polynomial (or the poles of t if the data is empty), keeps those of degree v, and keeps the one with matching key. It succeeds when the stored data singles out the place; the merge reports how many places do not satisfy this.

`X1_42_orbits_check.m` runs both reloading routes on every saved degree and prints PASS/FAIL lines.

7. Use in the divisor search

Let S be the sum of the 12 rational places (the cusps). Since G permutes the rational places, $\phi(S) = S$ for all $\phi \in G$. The manuscript (Corollary 3.3) reduces the statement $\mathrm{gon}_{\mathbb{F}_5}(X) \geq d+1$ to the following: for every effective divisor E of degree $d-2$ in a certain family $\mathcal{E}$, every nonconstant $h \in L(S+E)$ has degree $> d$. For $\phi \in G$, the map $h \mapsto h \circ \phi^{-1}$ is a degree-preserving bijection $L(S+E) \to L(S+\phi(E))$ (Lemma 4.1 of the manuscript). Hence it suffices to test one E from every G-orbit, provided $\mathcal{E}$ is G-stable.

Represent E by the multisets of indices of its places in each $\mathcal{P}_v$. Let K be the set of degrees for which π_v is known, and let E_{K} be the part of E supported in those degrees.

Proposition 7.1. *Order the representations of E_{K} lexicographically, and* keep *E if and only if $E_{\mathsf{K}} \leq \sigma^k(E_{\mathsf{K}})$ for $k = 1, \ldots, 11$. If $\mathcal{E}$ is G-stable, every G-orbit in $\mathcal{E}$ contains a kept divisor. If every degree occurring in divisors of $\mathcal{E}$ lies in K, every orbit contains exactly one kept divisor.*

Proof. Given E, choose k minimising $\sigma^k(E)_{\mathsf{K}} = \sigma^k(E_{\mathsf{K}})$, and put $E' = \sigma^k(E) \in \mathcal{E}$. For every j, $\sigma^j(E'_{\mathsf{K}}) = \sigma^{j+k}(E_{\mathsf{K}}) \geq E'_{\mathsf{K}}$, so E' is kept. Note that the images of places of degree outside K are never needed. If all degrees lie in K, then $E_{\mathsf{K}} = E$, and the lexicographic minimum of an orbit is unique. □

Degrees without a known permutation are thus simply not reduced, which is always safe. For $d = 11$ a test with synthetic permutations of the real place counts reduced 2,994,534 divisors to 531,211, a factor of about 5.6; the factor is below the group order 12 because divisors with nontrivial stabiliser have smaller orbits, and the parts of degree outside K are not reduced. The reduction is implemented in `X1_42_symreduce.py` (stage B). The search scripts (stage C) check, before searching, that the place numbering in the session agrees with the stored keys and checksum.

8. Status and extension

- *Tested commands.* `LocalUniformizer` returned a function with valuation 1 at all 20 test places (test T8b of `X1_42_symtest.m`). `UniformizingParameter` and `WeakApproximation([Q],[1])` are not available with these argument types in Magma V2.29-10, and are not used, except that `WeakApproximation` is tried first inside a `try`-block in the search scripts. Loading the group takes about 2 s, and the order check about 87 s.
- *To be confirmed on the server.* The output of `run_orbits.sh`, the merge summary, the PASS lines of `X1_42_orbits_check.m`, and test T11, which compares degrees 1–3 against A_4.
- *Higher degrees.* The method applies unchanged to $v = 6, \ldots, 9$; only more parts are needed. Computing $\mathcal{P}_9$ (with 214,384 places) is itself a substantial computation, repeated in every job. For these degrees, the place list and the data of Steps 1–2 should therefore be computed once and stored before the image jobs start.

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